Naman Goyal

Undergraduate Student
B.S. (Research) in Physics
Indian Institute of Science, Bengaluru

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Education

Indian Institute of Science — *Bengaluru, Karnataka*
B.S. (Research) in Physics — *Aug. 2023 – Expected May 2027*
Cumulative GPA: 9.4/10

Modern Vidya Niketan (MVN) — *Faridabad, Haryana*
Higher Secondary Schooling — *Apr. 2020 – Apr. 2023*
CBSE Class X: 97.8%
CBSE Class XII: 94.4%

Research Summary

My research interests lie in theoretical high energy physics and mathematical physics, with a particular focus on uncovering fundamental constraints imposed by quantum field theory and symmetry principles. My work combines analytical methods and scientific computing to study electroweak vacuum stability, semiclassical vacuum decay, scattering amplitudes, and effective descriptions of gravitation, with the broader goal of understanding the mathematical structure of fundamental interactions.

Research Experience

Tests of General Relativity in Pseudo-Newtonian Framework
Jan. 2024 – May 2026
Prof. Banibrata Mukhopadhyay (IISc) — *IISc, Bengaluru*

- Derived analytical expressions for perihelion precession, gravitational redshift, and gravitational lensing in several pseudo-Newtonian potentials, and compared their predictions with Schwarzschild General Relativity.
- Developed a Python framework for numerical analysis and visualization of orbital observables, generating the figures used to benchmark multiple pseudo-Newtonian models across different parameter regimes.
- Identified the regimes in which parameterized pseudo-Newtonian potentials successfully reproduce individual relativistic effects, while demonstrating that no single potential consistently captures perihelion precession, gravitational redshift, and gravitational lensing simultaneously.

Scattering Amplitudes and S-Matrix Bootstrap
May 2026 – Present
Prof. Rick Sandeepan Gupta (TIFR) — *TIFR, Mumbai*

- Investigating whether the uniqueness of dual-resonance scattering amplitudes can be established directly from analyticity, crossing symmetry, unitarity, and Regge behavior, avoiding auxiliary assumptions on their zero distribution.
- Studying analytic properties of scattering amplitudes through analyticity, crossing symmetry, unitarity, Regge behavior, and positivity constraints, drawing on modern developments in the S-matrix bootstrap.

Higgs Vacuum Stability and the Cosmological Hierarchy Problem [SMVac]
May 2024 – May 2026
Prof. Rick Sandeepan Gupta (TIFR) — *TIFR, Mumbai*

- Developed **SMVac**, a modular C++ framework for precision studies of electroweak vacuum stability, implementing NNLO matching conditions, 3-loop Standard Model renormalization group evolution, the one-loop Coleman–Weinberg effective potential, and semiclassical vacuum decay calculations.
- Computed Higgs vacuum decay rates using both the analytical Fubini–Lipatov formalism and exact numerical bounce solutions, constructing stability phase diagrams in the (M_h, M_t) plane and investigating the breakdown of analytical approximations near the metastability boundary.
- Investigated the Cosmological Hierarchy Problem by modeling the Higgs field as a stochastic spectator during slow-roll inflation. Studied whether volume-weighted anthropic selection during inflation can bias Higgs initial conditions toward metastable electroweak vacua without introducing new low-energy degrees of freedom.

Independent Reading: Lorentz Representations and Little Group Formalism
May 2025 – June 2025
Prof. Rick Sandeepan Gupta (TIFR) — *TIFR, Mumbai*

- Conducted a self-directed reading project on foundational quantum field theory, focusing on **Lorentz and Poincaré group representations** and the **little group formalism** for massive and massless particles.
- Investigated **continuous spin representations**, studying the formalism developed by Schuster and Toro, and analyzed their implications for exotic particle states beyond standard discrete spins.

Publications

1. **N. Goyal**, B. Mukhopadhyay, A. K. Meena. *Tests of General Relativity in Pseudo-Newtonian Approach*. arXiv:2602.20261 (2026).

Summer Schools & Academic Programs

Perimeter Institute Undergraduate Summer School (USS) *June 2026*
Perimeter Institute for Theoretical Physics, Waterloo, Canada

National Initiative on Undergraduate Science (NIUS) Physics Camp *June 2024*
Homi Bhabha Centre for Science Education (HBCSE–TIFR), Mumbai

Relevant Coursework

Advanced Theoretical Physics: Quantum Field Theory I & II, Standard Model of Particle Physics, General Relativity, Advanced Mathematical Physics, Nuclear and Particle Physics.
Core Physics & Astrophysics: Quantum Mechanics I & II, Classical Mechanics, Statistical Mechanics, Condensed Matter Physics, Fundamentals of Astrophysics, Electrodynamics, Thermal Physics.
Mathematics: Linear Algebra and Analysis, Probability & Statistics, Mathematical Methods of Physics.

Experimental Projects

Traveling Flame in Narrow Tubes
2024
International Physicists' Tournament (IPT) *IISc, Bengaluru*

- Designed and 3D-printed narrow tubes with controlled obstacles to study flame propagation dynamics for the International Physicists' Tournament.
- Conducted experiments characterizing ignition thresholds, flame speed regimes, and the transition from laminar deflagration to turbulent combustion.
- Analyzed high-speed footage and pressure data to quantify turbulence onset, providing insights into premixed flame behavior in confined geometries.

Jumping Hoop Phenomenon
2023
International Physicists' Tournament (IPT) *IISc, Bengaluru*

- Developed a theoretical model for the vertical jumping of a rigid hoop driven by an off-center sliding puck, incorporating rotational inertia, impulsive normal forces, and shifting center of mass.
- Performed controlled experiments measuring jump heights and trajectories under varying mass ratios and initial conditions.
- Implemented numerical simulations in C++ to solve the coupled equations of motion, achieving good agreement with experimental observations.

Technical Skills

Programming Languages	C++, Python
Scientific Computing	Adaptive ODE integration, Multi-loop RGE evolution, Numerical optimization, Scientific visualizatio
Libraries & Software	NumPy, SciPy, Matplotlib, SymPy, Mathematica
Development Tools	Git, LaTeX

Awards and Honors

Joint Entrance Examination (JEE) 2023
Secured an **All India Rank of 33** in JEE Advanced (among ∼180,000 candidates) and an **AIR 281** in JEE Main (among ∼1.1 million candidates).

PLANCKS 2024 – International Physics Competition
Represented India at the International Association of Physics Students (IAPS) PLANCKS finals in Dublin, Ireland, achieving an **11th-place global finish**. Re-selected to represent India in the 2025 competition.

National Initiative on Undergraduate Science (NIUS) Physics Scholar
Selected by HBCSE–TIFR as a NIUS Physics Scholar (Batch 21.1), a national program admitting approximately **70 undergraduate students** annually for advanced research training.

Indian Physics Tournament (IPT)
Member of the IISc team that secured **National Runner-up** in both the 2023 and 2024 editions through contributions to theoretical modeling and experimental design.

Kishore Vaigyanik Protsahan Yojana (KVPY) Scholar
Awarded the KVPY fellowship with an **AIR 171** (SA Stream), a national scholarship supporting students pursuing research careers in the basic sciences.

Indian Astronomy Olympiad (InAO)
Selected twice (2021, 2022) for the Orientation-cum-Selection Camp (OCSC), the final national selection stage for India's International Olympiad in Astronomy and Astrophysics team.

Regional Mathematics Olympiad (RMO) & National Standard Examination in Physics (NSEP)
Achieved Top 1% state ranking in RMO (2020, 2021) and Top 1% state ranking in NSEP (2020).